

Kiran Kuruvila

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Professional Summary

Machine Learning Engineer focused on building AI systems, quantitative pipelines, and computer-vision based automation workflows. Experienced in transformers, deep learning architectures, agentic LLM systems, RAG and real-time data pipelines. Hands-on experience deploying containerized applications on AWS infrastructure using Docker, EC2, Nginx, Linux, and Amazon ECR.

Education

Marian College, Kuttikkanam, Kerala <i>Master of Computer Applications (MCA)</i>	2025 – 2027
MG University, Kottayam, Kerala <i>Bachelor of Arts in Economics</i>	2022 – 2025

Projects

Agentic Trading System | *Python, LangGraph, Pydantic*

- Designed a modular multi-agent workflow system using LangGraph for financial signal processing and decision orchestration
- Implemented typed state management and validation pipelines using Pydantic for deterministic execution reliability
- Built structured data ingestion and feature-engineering workflows for quantitative analysis pipelines

Quantaire: Stock Market Pattern Detection | *Python, YOLOv8, Playwright, NumPy, Docker, AWS ECR*

- Built a computer-vision pipeline for automated detection of technical stock-chart patterns using a custom YOLOv8 model
- Automated TradingView chart capture using Playwright with headless browser orchestration and retry-based session recovery
- Containerized inference services using Docker and deployed images through Amazon Elastic Container Registry (ECR)

GPT-Style Transformer for Algorithmic Reasoning | *PyTorch, NumPy*

- Built a decoder-only Transformer model for arithmetic sequence prediction tasks
- Designed synthetic datasets and preprocessing pipelines for supervised learning workflows
- Implemented evaluation and optimization workflows for training stability and performance analysis

NanoGPT: Character-Level Language Model | *PyTorch*

- Implemented a GPT-style architecture with causal self-attention and token embedding workflows
- Trained language models on structured text datasets using AdamW optimization and learning-rate scheduling

Gap Scanner Telegram Bot | *Python, yfinance, python-telegram-bot, Cron*

- Built an automated market gap scanner that fetches live stock data daily at 9:17 AM via a cron job and analyzes price gaps across a watchlist
- Implemented gap detection logic to flag stocks exceeding $\pm 1\%$ threshold, filtering actionable signals from noise
- Delivered structured gap reports to a private Telegram group using the Telegram Bot API, restricted to authorized members

Cloud Deployment & Backend Infrastructure | *Django, MySQL, AWS, Linux, Docker, Nginx*

- Deployed containerized applications on AWS EC2 instances with Nginx reverse proxy configuration
- Worked with Linux server environments, deployment pipelines, process management, and remote infrastructure administration
- Managed relational database workflows using MySQL and AWS RDS

Recruitment Platform | *Django, MySQL, AWS EC2, Linux, Nginx, Gunicorn*

- Built a recruitment platform backend using Django with JWT-based authentication and MySQL
- Implemented job and application management functionality for job seekers and employers
- Deployed the application on AWS EC2 using Gunicorn and Nginx

Neural Network Foundations | *Python, PyTorch*

- Implemented deep learning architectures from first principles including autograd engines, RNNs, Transformers, and CNNs

Teaching & Knowledge Sharing

Neural Networks Fundamentals Session

- Conducted peer-level sessions explaining neural network fundamentals from first principles
- Taught gradient descent, backpropagation, optimization dynamics, and neural computation using mathematical intuition

Technical Skills

Languages: Python, SQL, Java

Backend Frameworks: Django, FastAPI, REST APIs

ML / AI: PyTorch, Transformers, CNNs, RNNs (GRUs, LSTM), Reinforcement Learning, LangGraph, RAG

Data & ML Libraries: NumPy, Pandas, scikit-learn

Cloud & DevOps: AWS (EC2, RDS, ECR, EFS), Docker, Gunicorn, Linux, Nginx, Git

Concepts: OOP, JWT Authentication, RESTful API Design, Feature Engineering, Agent Workflows, Data Pipelines, Model Deployment

Mathematics: Probability, Statistics, Linear Algebra, Calculus